

Stellar Subdivision Schemes on Balanced Triangulations: Pointerless Maubach Bisection and Riemann Surfaces in $\mathbb{CP}^2$

Vinícius Moreira Mello

Departamento de Matemática, Universidade Federal da Bahia, Salvador, Brazil

vinicius.mello@ufba.br

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Abstract

We revisit newest-vertex (Maubach) bisection from a purely combinatorial standpoint. Following the framework of *stellar subdivision schemes*, a scheme is a pair formed by a combinatorial invariant on ordered simplicial meshes and a subdivision algorithm that provably preserves it, stated and proved level by level through ordered stellar operations alone. In this language, the facet-based scheme of Mitchell and the edge-based scheme of Maubach have the same deductive structure. We then study which *initial* meshes a scheme can start from. For Maubach’s scheme, the strong form of the invariant on a conforming mesh is equivalent to the mesh being *balanced*, i.e. properly $(n+1)$ -colorable. This places the classical constructions (products of simplices, barycentric subdivisions) and the recent coloring-based initializations in a single picture, and it applies verbatim to triangulations of *closed manifolds*, a setting absent from the finite-element literature. As a case study we show that Gaifullin’s 15-vertex triangulation of $\mathbb{CP}^2$ is balanced, and hence, by Gaifullin’s minimality theorem, a vertex-minimal balanced triangulation of $\mathbb{CP}^2$, while Kühnel’s 9-vertex $\mathbb{CP}_9^2$ is not. Balance is also exactly what makes Atalay and Mount’s pointerless LPT codes portable. Every seed cell, ordered by color, is a copy of the Kuhn reference simplex. Whether a facet lies on the seed boundary is decided by one table lookup and one integer comparison, and the neighbor across it is the same code in the adjacent seed cell. A seed index and a seed adjacency table therefore suffice to run pointerless, conforming Maubach bisection over any balanced seed, with no floating point. We illustrate the whole construction by triangulating smooth plane algebraic curves, i.e. compact Riemann surfaces, as codimension-2 loci *intrinsically* inside $\mathbb{CP}^2$, by a continuation algorithm driven entirely by the combinatorial neighbor relations of the pointerless mesh.

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1 Introduction

Adaptive refinement of simplicial meshes by *bisection* is one of the oldest tools of computational geometry and numerical analysis. Among the many variants, newest-vertex bisection in the form given by Maubach [39] and Traxler [54] stands out in arbitrary dimension n . It is *blind*, meaning the bisection edge of a cell is determined by combinatorial data alone. It is conforming and produces at most $n n! 2^{n-2}$ similarity classes per initial simplex [4]. It also admits optimal closure estimates [10, 53]. Its one well-known weakness is the initial mesh: the method only works if the vertices of the initial cells are ordered *compatibly*. For $n \geq 3$ such orderings do not exist for every mesh, and a substantial literature addresses how to repair this [34, 53, 1, 8, 19].

This paper has three strands.

1. Stellar subdivision schemes. The author’s thesis [43], available only in Portuguese, developed a framework in which a *subdivision scheme* is a pair: a set of combinatorial invariants on ordered simplicial meshes, and an algorithm $\text{Subdivide}(K, \sigma)$ that propagates ordered stellar subdivisions until σ can be split without violating the invariants. Correctness is proved by a short induction on the hierarchy level, using only the face operators ∂_i of the ordered mesh. The framework covers Mitchell’s facet-based scheme [45] and Maubach’s edge-based scheme with the same deductive skeleton. We present it here in English, in a self-contained way (Sections 3 and 4).

2. Initial meshes. We show that, on conforming meshes, the strong form of Maubach’s invariant (“every shared facet sits at the same local index on both sides”) is equivalent to the mesh being *balanced* in the sense of Stanley, i.e. admitting a proper vertex coloring by $n + 1$ colors (Proposition 5.2). The thesis’s two constructions, the canonical triangulation of a product of simplices and the barycentric subdivision, become one-line corollaries. So does a product

construction for already-balanced meshes. This is the $N = n$ case of the recent coloring-based initialization of Diening, Gehring and Storn [19], and is equivalent to Traxler’s reflected-partition condition [54]. What seems new is its use on *closed-manifold seeds*. We prove that Gaifullin’s 15-vertex triangulation of $\mathbb{CP}^2$ [26] is balanced, and deduce from Gaifullin’s own minimality theorem that no balanced triangulation of $\mathbb{CP}^2$ has fewer vertices (Theorem 5.10). Kühnel’s 9-vertex $\mathbb{CP}_9^2$ [36] is not balanced (Proposition 5.9).

3. Pointerless codes. Atalay and Mount’s LPT codes [5] represent a Maubach hierarchy over a Kuhn-triangulated cube with no pointers at all. Geometry and neighbors of a cell are computed from a small integer code. We show that the only feature of the cube their neighbor rules really use is balance. Consequently, the codes extend to any balanced seed mesh by adding a seed index and a seed adjacency table. The seed-boundary test reduces to a precomputed table plus one comparison of orthant coordinates, and crossing into the next seed cell keeps the code unchanged (Section 6). We also quantify why codes matter: the storage of an explicit incidence structure grows roughly by a factor 1.6 per dimension, and it is already 11 times that of the codes for $d = 4$.

Application. We combine the three strands to triangulate a smooth projective plane curve $\{F = 0\} \subset \mathbb{CP}^2$, a compact Riemann surface, directly as a codimension-2 submanifold of the real 4-manifold $\mathbb{CP}^2$. We seed with Gaifullin’s triangulation, refine by pointerless Maubach bisection along Fubini–Study geodesics, and follow the curve by a continuation algorithm that uses only the mesh’s combinatorial neighbor queries (Section 7). Unlike chart- or branch-cut-based approaches, the construction is intrinsic: points at infinity play no special role.

Contributions.

1. An English, self-contained account of stellar subdivision schemes and of Maubach’s scheme within them. It includes an invariant that is preserved at every level of the hierarchy, and its precise relation to Stevenson’s matching condition (Sections 3 and 4).
2. The equivalence “strongly Maubach-compatible $\iff$ balanced” for conforming meshes, a counterexample showing that index-only matching is insufficient, and closure of balance under products (Section 5).
3. Balanced seeds for closed manifolds. Gaifullin’s triangulation is a vertex-minimal balanced triangulation of $\mathbb{CP}^2$ (Theorem 5.10).
4. An extension of LPT codes to arbitrary balanced seeds. It has a fully combinatorial neighbor procedure: an $O(1)$ table-based seed-boundary test and same-code seed crossing. It comes with a graded, conforming leaf-set data structure and a quantitative comparison with explicit incidence storage (Section 6).
5. An intrinsic, adaptive triangulation algorithm for Riemann surfaces in $\mathbb{CP}^2$ based on combinatorial continuation, with measurements (Section 7).

Section 2 reviews related work and states how our approach compares.

2 Related work

Bisection and initial meshes. Longest-edge bisection goes back to Rivara [51, 52]. Newest-vertex bisection (NVB) was introduced in 2D by Mitchell [44, 45], extended to 3D by Bänsch [6] and Kossaczky [34], and to arbitrary dimension by Maubach [39] and Traxler [54]. Arnold, Mukherjee and Pouly [4] proposed a marked variant for tetrahedra that accepts any initial

mesh. Binev, Dahmen and DeVore [10] and Stevenson [53] proved the closure estimates that underlie optimality of adaptive finite elements. All of these methods constrain the initial mesh.

- Traxler’s *reflected* condition requires the ordered vertex lists of neighbors to differ in exactly one position. On a simply connected domain it holds iff every interior $(n-2)$ -face has even degree.
- Stevenson’s weaker *matching-neighbour* condition is necessary and sufficient for all uniform refinements to be conforming. Whether every 3D mesh admits it is open. Following Kossaczky, Stevenson therefore pre-refines every simplex into $\frac{1}{2}(n+1)!$ pieces.
- Alkämper, Gaspoz and Klöfkorn [1] give an $O(n)$ renumbering guaranteeing termination.
- Belda-Ferrín et al. [7, 9, 8] perform n marked-bisection stages before switching to Maubach, and note that for $n > 3$ “there is not any known method to extract a reflection structure from an arbitrary unstructured conformal mesh”.
- Diening, Gehring and Storn [19] initialize Maubach’s routine from a proper $(N+1)$ -coloring of the vertices, with $N \geq n$, computed greedily in linear time. When $N > n$ each cell is treated as a face of a virtual N -simplex. They prove shape regularity and the closure estimate. For $N = n$ their condition is Traxler’s; see also [20, 27].

All of the above concern triangulations of domains in $\mathbb{R}^n$.

Our treatment differs from this literature in two respects. First, the compatibility condition is formulated as an *invariant*, hereditary in the level, rather than as a condition on the initial mesh only. Second, we apply it to seeds that triangulate closed manifolds. The equivalence of Proposition 5.2 makes the connection with the coloring viewpoint of [54, 19] explicit.

Stellar theory and balanced complexes. Stellar subdivisions and welds go back to Alexander and Newman; see [38] for a modern account. In geometry processing, Velho’s stellar subdivision grammars [55] and 4–8 meshes [56] describe adaptive surface refinement by stellar operations; 4–8 refinement is NVB in two dimensions. A d -complex is *balanced* if its graph is $(d+1)$ -colorable. Joswig [32] characterized balanced complexes through their group of projectivities. Izmetiev, Klee and Novik [31] introduced cross-flips, the balanced analogue of bistellar moves. Venturello [57] found small balanced triangulations of several manifolds. We are not aware of work connecting balanced triangulations of closed manifolds with bisection refinement.

Pointerless hierarchical meshes. Several representations encode a simplex of a regular bisection hierarchy by a label rather than by pointers:

- Hebert [30], Evans, Kirkpatrick and Townsend [25], and Lee, De Floriani and Samet [37], in 2D and 3D;
- Maubach’s own neighbor location scheme [40];
- Atalay and Mount’s LPT codes [5], in arbitrary dimension over a Kuhn-triangulated cube, which we previously re-implemented to generate adaptive cellular infill structures for 3D printing [23];
- Weiss and De Floriani’s diamond hierarchies [59], which use the star of the bisection edge as primitive;
- Dupuy’s concurrent binary trees [21], which implement longest-edge bisection on the GPU;
- Boissonnat, Kachanovich and Wintraecken’s permutahedral encoding of Freudenthal–Kuhn simplices [12], which is compact but not adaptive.

The closest architectural analogue of our construction is the *forest* approach of p4est [16] and t8code [15]. There, an arbitrary coarse mesh is refined tree by tree, with a per-tree linear code and an inter-tree connectivity table. Those forests are *non-conforming*: they allow hanging nodes and use red refinement in 2D and 3D. Ours is a conforming, graded bisection forest in arbitrary dimension.

Riemann surfaces and implicit manifolds. Numerical approaches to compact Riemann surfaces work with a branched covering of $\mathbb{CP}^1$ and compute analytic invariants such as monodromy and period matrices [18, 11, 35]. Visualizations use parametrizations [29] or multivalued graphs over $\mathbb{C}$ with branch cuts [47]. Our construction is a simplicial complex of the curve inside $\mathbb{CP}^2$ itself.

Piecewise-linear approximation of implicit manifolds of arbitrary codimension over Freudenthal–Kuhn triangulations goes back to Allgower and Schmidt [3, 2]. Other related constructions are:

- Weigle and Banks [58] extract the zero set of a function $\mathbb{C}^2 \rightarrow \mathbb{C}$ as a surface in $\mathbb{R}^4$ on a uniform grid. This is the closest predecessor of our codimension-2 extraction, but it works in an affine chart.
- Castelo et al. [17] give an adaptive triangulation in any dimension for implicit manifolds.
- Boissonnat, Kachanovich and Wintraecken [12] trace isomanifolds cell by cell from a seed point over uniform Coxeter–Freudenthal–Kuhn triangulations, with guarantees of topological correctness [13]. Our continuation follows the same strategy, over an adaptive mesh of a compact manifold, but without certification.
- Certified alternatives exist for curves and surfaces in low dimension: interval-based subdivision [49], Bernstein subdivision [46, 50], and numerical cell decomposition [14].
- Bisection hierarchies are also used for isosurface extraction in $\mathbb{R}^3$ [28, 48, 33].

3 Ordered meshes and stellar subdivision schemes

Adaptive refinement has two sides. One is geometric: where to put the new vertices and how cell shapes evolve. The other is combinatorial: which simplex to split next, and which additional splits keep the mesh conforming. Most bisection methods mix the two, for example by choosing the longest edge. Newest-vertex bisection is attractive precisely because its decisions can be made from combinatorial data alone. That is what makes it usable on meshes whose geometry is unusual, such as a triangulation of $\mathbb{CP}^2$ realized with Fubini–Study geodesics, where “longest edge” has no convenient meaning. The goal of this section is to isolate that combinatorial layer.

We work with *ordered* simplicial complexes, in which the vertex order of every simplex is part of the data. Refinement becomes a sequence of *ordered stellar subdivisions*. A refinement *scheme* is then specified by an invariant, a property of the order and of the hierarchy levels, together with an algorithm that preserves it. Correctness proofs reduce to checking, locally, how a single stellar operation affects pairs of adjacent cells. The Mitchell and Maubach schemes differ only in their invariants. The same three lemmas, and the same induction on the hierarchy level, prove both.

3.1 Ordered simplicial complexes and conforming meshes

Definition 3.1. An *ordered simplicial complex* K on a finite vertex set V is an abstract simplicial complex together with a partial order $<$ on V such that the vertices of every simplex are totally ordered. We write $\sigma = \langle v_0, \dots, v_n \rangle$ with $v_0 < \dots < v_n$ and $\sigma[j] = v_j$. The *face operators* $\partial_i : K_n \rightarrow K_{n-1}$ delete the i -th vertex. They satisfy $\partial_j \partial_i = \partial_{i-1} \partial_j$ for $j < i$. More generally, for $0 \leq j_0 < \dots < j_k \leq n$ we write $\sigma[j_0, \dots, j_k] := \langle v_{j_0}, \dots, v_{j_k} \rangle$.

Equalities such as $\partial_i \sigma = \partial_j \omega$ are equalities of *ordered* simplices. This will matter in Remark 5.3. The order encodes, locally, everything the subdivision algorithms need. No geometric information is ever consulted.

For an m -dimensional complex, m -simplices are *cells* and $(m-1)$ -simplices are *facets*. K is a *pseudomanifold* if it is pure and every facet lies in at most two cells. The *dual graph* $G(K)$ has the cells as nodes and the facets shared by two cells as edges. K is *strongly connected* if $G(K)$ is connected.

Definition 3.2 ([43]). A strongly connected pseudomanifold K is *hereditary* if the star $\text{st}(v, K)$ of every vertex is strongly connected. A hereditary pseudomanifold is called a *conforming mesh*.

In a conforming mesh the star of every simplex is strongly connected. The star of any face can therefore be traversed by walking across facets, which is how the algorithms below find cells.

3.2 Ordered stellar subdivisions

Let $\delta \in K$ and $v \notin V$. The *stellar subdivision* $K' = K(\delta, v)$ replaces $\text{st}(\delta, K)$ by $v \star \partial \delta \star \text{link}(\delta, K)$. We write $K' = K(\delta, v)_l$, an *ordered stellar subdivision*, when the order on $V \cup \{v\}$ places v at position l in every cell of $\text{st}(v, K')$.

Let σ be an m -cell and $\delta = \sigma[j_0, \dots, j_n]$ one of its faces. Applying $(\delta, v)_l$ to σ yields $n+1$ children $c_0 \sigma, \dots, c_n \sigma$, one for each facet of $\partial \delta$. They can be indexed so that

$$\partial_l c_i \sigma = \partial_{j_{n-i}} \sigma, \quad (1)$$

and orientations then propagate by $o(c_i \sigma) = o(\sigma)(-1)^{l+j_{n-i}}$.

A sequence of ordered stellar subdivisions starting from a mesh K induces a hierarchy on cells. Let $\text{level}(\sigma) = 0$ for $\sigma \in K$ and $\text{level}(c_i \sigma) = \text{level}(\sigma) + 1$. Every cell σ of positive level has a unique *parent* $p(\sigma)$ and a unique *weld vertex* $\text{weld}(\sigma) \in \sigma \setminus p(\sigma)$, the vertex whose removal by a stellar weld would undo the subdivision that created σ .

3.3 Subdivision schemes

Definition 3.3 ([43]). A *stellar subdivision scheme* is a pair $(\mathcal{I}, \text{Subdivide})$. Here $\mathcal{I}$ is a set of invariants referring only to combinatorial data of a conforming mesh: vertex order, adjacency, level, and possibly labels with prescribed behavior under subdivision. Given a mesh K satisfying $\mathcal{I}$ and a cell σ , the algorithm $\text{Subdivide}(K, \sigma)$ applies a sequence of ordered star subdivisions $(\delta_1, v_1)_{l_1} \cdots (\delta_q, v_q)_{l_q}$ in which only δ_q , the *subdivision face*, is a face of σ . The result, and every intermediate mesh, must again satisfy $\mathcal{I}$.

A typical invariant bounds the level jump across facets:

$$\partial_i \sigma = \partial_j \omega \implies |\text{level}(\sigma) - \text{level}(\omega)| \leq 1. \quad (\text{OneLevel})$$

The thesis names three desirable properties of Subdivide . To avoid a clash with *balanced complexes* (Section 5), we rename the third property here. Subdivide is:

- *consistent*, if there is a function split assigning each cell σ' a face $\text{split}(\sigma') \subset \sigma'$ such that every cell in the star of a subdivided face δ has $\text{split}(\sigma') = \delta$. All cells split in one stellar operation agree on what to split, and a generic weld algorithm can undo the refinement;
- *alternating*, if $\text{weld}(\sigma) \notin \text{split}(\sigma)$ for every cell. Newly created faces are never immediately split, which forces propagation outward;
- *level-homogeneous* (“balanced” in [43]), if all cells in the star of a subdivided face have the same level.

3.4 Mitchell's scheme

Mitchell's scheme always splits the facet opposite the first vertex, $\text{split}(\sigma) := \partial_0\sigma$. Written as invariants, this becomes the following definition.

Definition 3.4 (Mitchell invariants). K satisfies the *Mitchell invariants* if it satisfies (OneLevel) and, for adjacent cells with $\partial_i\sigma = \partial_j\omega$,

$$\text{level}(\sigma) = \text{level}(\omega) \implies i = 0 = j \text{ or } i \neq 0 \neq j, \quad (\text{MitchellA})$$

$$\text{level}(\sigma) = \text{level}(\omega) + 1 \implies i = 0 \neq j. \quad (\text{MitchellB})$$

(MitchellA) says two same-level neighbors either share their split facet or neither contains the other's. (MitchellB) says that splitting the coarser cell hands the split facet to one of its children. Any mesh satisfies the invariants after the ordered cone $(\sigma, v_\sigma)_0$ is applied to every cell. The algorithm is two lines: if the neighbor across $\partial_0\sigma$ is coarser, subdivide it first; then split $\partial_0\sigma$ at position 0.

Theorem 3.5 ([43]). *Mitchell's algorithm preserves the Mitchell invariants and is consistent, alternating and level-homogeneous.*

The proof uses three lemmas that recur, in the same roles, for Maubach's scheme:

- a *direction lemma* saying which neighbor can be coarser;
- a *child lemma* describing the adjacencies among the children of a single split cell;
- a *two-cell lemma* showing the invariants survive splitting one of two adjacent cells.

The induction on $\text{level}(\sigma)$ then assembles them. We give the Maubach versions in full below. The Mitchell versions are simpler.

4 Maubach's scheme

In dimension 2 a facet is an edge, and Mitchell's scheme is exactly newest-vertex bisection. In higher dimensions, however, splitting facets gives no control over the shape of the cells. Maubach's scheme instead splits an edge. On Kuhn's triangulation of the cube it reproduces longest-edge bisection and produces only finitely many similarity classes. The question for this section is what Maubach's rule looks like as a stellar subdivision scheme: which combinatorial invariant on an ordered mesh guarantees that the edge chosen by one cell is also the edge chosen by every neighbor sharing it. Answering it gives a proof of conformity that refers only to the vertex order and the levels, and so applies unchanged to any mesh satisfying the invariant, whatever its geometry.

Maubach's scheme splits an *edge*. For an m -dimensional mesh set

$$\text{type}(\sigma) := m - (\text{level}(\sigma) \bmod m) \in \{1, \dots, m\}, \quad \text{split}(\sigma) := \sigma[0, \text{type}(\sigma)].$$

The new vertex is inserted at position $\text{type}(\sigma)$. By (1), the two children of $\sigma = \langle v_0, \dots, v_m \rangle$ under $(\sigma[0, l], v)_l$, with $l = \text{type}(\sigma)$, are

$$c_0\sigma = \langle v_0, \dots, v_{l-1}, v, v_{l+1}, \dots, v_m \rangle, \quad c_1\sigma = \langle v_1, \dots, v_l, v, v_{l+1}, \dots, v_m \rangle.$$

This is exactly the rule of [39], and of [19, Alg. 1].

Definition 4.1 (Maubach invariants). The mesh K satisfies the *Maubach invariants* if it satisfies (OneLevel) and, for adjacent cells with $\partial_i\sigma = \partial_j\omega$,

$$\text{level}(\sigma) = \text{level}(\omega) \implies i = j \text{ or } \{i, j\} = \{0, \text{type}(\omega)\}, \quad (\text{MaubachA})$$

$$\text{level}(\sigma) = \text{level}(\omega) + 1 \implies i = \text{type}(\omega) \text{ and } j \in \{0, \text{type}(\omega)\}. \quad (\text{MaubachB})$$

We say K is *strongly compatible* if $\partial_i\sigma = \partial_j\omega \implies i = j$ for all adjacent cells. At level 0 this implies the Maubach invariants.

(MaubachA) implies that two same-level neighbors either share their split edge or neither contains the other's. Indeed, if $\text{split}(\sigma) \subset \omega$ then $i \notin \{0, \text{type}\}$, so $i = j$ and the two cells differ in a single vertex away from the split edge. (MaubachB) implies that splitting the coarser cell gives one of its children exactly the facet the finer neighbor already has.

Lemma 4.2 (direction). *If K satisfies the Maubach invariants and $\partial_i \sigma = \partial_j \omega$, then $\text{level}(\omega) \leq \text{level}(\sigma)$ when $i \notin \{0, \text{type}(\sigma)\}$, and $\text{level}(\omega) \geq \text{level}(\sigma)$ when $i \in \{0, \text{type}(\sigma)\}$. Moreover, $\text{level}(\omega) < \text{level}(\sigma)$ implies $i = \text{type}(\omega)$.*

Proof. If $\text{level}(\omega) > \text{level}(\sigma)$ then $\text{level}(\omega) = \text{level}(\sigma) + 1$ by (OneLevel). (MaubachB), with the roles of the cells exchanged, gives $i \in \{0, \text{type}(\sigma)\}$. If $\text{level}(\omega) < \text{level}(\sigma)$, then (MaubachB) gives $i = \text{type}(\omega)$. Since $\text{type}(\omega) \neq \text{type}(\sigma)$ and $\text{type} \neq 0$, it follows that $i \notin \{0, \text{type}(\sigma)\}$. $\square$

Lemma 4.3 (children). *For a single cell ω and $l > 0$, the children of $(\omega[0, l], v)_l$ satisfy $\partial_0 c_0 \omega = \partial_{l-1} c_1 \omega$.*

Proof. Both sides equal $\langle v_1, \dots, v_{l-1}, v, v_{l+1}, \dots, v_m \rangle$. $\square$

Lemma 4.4 (two cells). *Let K consist of two adjacent cells with $\partial_i \sigma = \partial_j \omega$ and $0 \leq \text{level}(\sigma) - \text{level}(\omega) \leq 1$, satisfying the Maubach invariants. Then $K(\omega[0, l], v)_l$, with $l = \text{type}(\omega)$, satisfies them as well.*

Proof. There are seven cases. They are distinguished by whether the levels are equal or differ by one, and by which of i, j lie in $\{0, l\}$. In each case, Lemma 4.3 settles the pair $c_0 \omega, c_1 \omega$. The remaining relation involving σ reduces to a single facet identity:

	levels	indices	resulting relation	
1	equal	$i = j = 0$	$\partial_0 \sigma = \partial_l c_1 \omega$	(MaubachB)
2	equal	$i = j = l$	$\partial_l \sigma = \partial_l c_0 \omega$	(MaubachB)
3	equal	$i = j = k \notin \{0, l\}$	σ is split too; $\partial_k c_0 \sigma = \partial_k c_0 \omega$, $\partial_{k-1} c_1 \sigma = \partial_{k-1} c_1 \omega$	(MaubachA)
4	equal	$i = 0, j = l$	$\partial_0 \sigma = \partial_l c_0 \omega$	(MaubachB)
5	equal	$i = l, j = 0$	$\partial_l \sigma = \partial_l c_1 \omega$	(MaubachB)
6	+1	$i = l, j = 0$	$\partial_l \sigma = \partial_l c_1 \omega$	(MaubachA)
7	+1	$i = l, j = l$	$\partial_l \sigma = \partial_l c_0 \omega$	(MaubachA)

In cases 1–5, σ becomes one level coarser than the children of ω , and (MaubachB) applies. In cases 6–7 the levels become equal and (MaubachA) applies. $\square$

Theorem 4.5 ([43]). *If K satisfies the Maubach invariants, so does the result of Algorithm 1 applied to any cell. The algorithm is consistent, alternating and level-homogeneous.*

Proof. The proof is by induction on $\text{level}(\sigma)$. Suppose σ has the minimal level in K . By Lemma 4.2, every cell sharing the split edge ϵ has the same level, so the recursive call never fires. The loop then walks $\text{st}(\epsilon, K)$, which is possible because K is conforming. By (MaubachA) every cell in the star has split edge ϵ , which gives consistency and level-homogeneity. Lemma 4.4, applied to every adjacent pair meeting the star, preserves the invariants. For higher levels, Lemma 4.2 and the induction hypothesis applied to the recursive call reduce to the base case. $\square$

4.1 Relation to Traxler and Stevenson

Traxler [54] and Stevenson [53] constrain only the initial mesh, and then prove that all uniform refinements are conforming [53, Thm. 4.3]. Stevenson uses Traxler's convention. A tagged simplex $(x_0, \dots, x_n)_\gamma$ is bisected along $x_0 x_n$, and $T_R = (x_n, x_1, \dots, x_\gamma, x_{n-1}, \dots, x_{\gamma+1}, x_0)_\gamma$ denotes the tagged simplex with the same children. Two neighbors are *reflected* if the vertex sequence of T or of T_R agrees with that of T' in all but one position. His condition (b) on the initial mesh has two clauses:

Algorithm 1 MAUBACHSUBDIVIDE(K, σ) [43]

Require: conforming mesh K satisfying the Maubach invariants; cell σ

```
1:  $k \leftarrow \text{type}(\sigma)$ ;  $st \leftarrow []$ ;  $Q \leftarrow (\sigma)$ 
2: while  $Q \neq \emptyset$  do
3:    $c \leftarrow \text{pop}(Q)$ 
4:   if  $c$  not visited then
5:     mark  $c$  visited; append  $c$  to  $st$ 
6:     for  $i \in \{0, \dots, m\} \setminus \{0, k\}$  do ▷ facets containing the split edge
7:        $c_a \leftarrow \text{adjacent}(c, i)$ 
8:       if  $c_a$  not visited then
9:         if  $\text{level}(c_a) < \text{level}(c)$  then
10:           MAUBACHSUBDIVIDE( $K, c_a$ );  $c_a \leftarrow \text{adjacent}(c, i)$ 
11:         end if
12:         push  $c_a$  onto  $Q$ 
13:       end if
14:     end for
15:   end if
16: end while
17: split the edge  $\sigma[0, k]$  at position  $k$  in every cell of  $st = \text{st}(\sigma[0, k], K)$ 
```

- if the refinement edge of either cell lies on the common facet, the two cells are reflected neighbors;
- otherwise, their neighboring children are.

At level 0, Traxler's type $\gamma = 0$ with a given vertex order corresponds to Maubach's type n with the same order [19, Rem. 4]. In both cases the refinement edge is the first–last pair.

Proposition 4.6. *Let K be a level-0 mesh satisfying (MaubachA). For adjacent cells σ, ω :*

- (a) *if $\partial_i \sigma = \partial_i \omega$, their vertex sequences agree in all but position i ;*
- (b) *if $\partial_0 \sigma = \partial_n \omega$, neither refinement edge lies on the common facet, and the children containing the common facet have vertex sequences that agree in all but one position.*

Hence (MaubachA) at level 0 implies the analogue of Stevenson's condition (b), with T always used in place of T_R .

Proof. (a) is immediate from the definition. For (b), write $\sigma = \langle a, f_1, \dots, f_n \rangle$ and $\omega = \langle f_1, \dots, f_n, b \rangle$. The refinement edges are $\{a, f_n\}$ and $\{f_1, b\}$, and neither lies in $\{f_1, \dots, f_n\}$. By Lemma 4.3 with $l = n$, the children containing $\{f_1, \dots, f_n\}$ are $c_1 \sigma = \langle f_1, \dots, f_n, v_\sigma \rangle$ and $c_0 \omega = \langle f_1, \dots, f_n, v_\omega \rangle$. They differ only in the last position. $\square$

The converse fails for representational reasons. Stevenson's clause using T_R relates neighbors whose common vertices appear in *reversed* order. Equalities of ordered faces cannot express this, so each cell must commit to one of T and T_R . Stevenson also proves that condition (b) is necessary for conformity [53, Rem. 4.4]. Together with Theorem 4.5, this shows that (MaubachA) at level 0 lies between Traxler's condition and Stevenson's. The structural novelty of Definition 4.1 is that it is *hereditary*: it holds at every level, which is what makes the local, inductive proof of Theorem 4.5 possible.

Remark 4.7 (Geometry). The scheme is purely combinatorial. Where the new vertex is placed is a separate choice. Midpoint placement on a Kuhn cube reproduces Maubach's geometric theorem: cells of equal type are similar [39]. The thesis gives a short proof via the combinatorial type. In Section 7 we place vertices at *Fubini–Study geodesic* midpoints instead, and the combinatorics is unaffected.

5 Initial meshes: strong compatibility is balance

Theorem 4.5 moves the whole difficulty to the initial mesh: once the Maubach invariants hold at level 0, refinement takes care of itself. For a product of simplices, such as Kuhn’s cube, they hold by construction. In general they do not, and this is the classical obstacle to newest-vertex bisection in dimension $n \geq 3$. Our application raises the question in a setting the finite-element literature does not consider: the seed is a triangulation of a closed 4-manifold, $\mathbb{CP}^2$, rather than of a domain in $\mathbb{R}^n$. Which triangulations of $\mathbb{CP}^2$ can serve as seeds, and how can we recognize them?

This section shows that the strong form of the invariant is a familiar combinatorial property in disguise: the mesh must be *balanced*, i.e. its vertices must admit a proper coloring by $n + 1$ colors. This reduces the question to graph coloring and connects it with the literature on balanced manifolds. It also explains the classical constructions uniformly, and it singles out Gaifullin’s triangulation of $\mathbb{CP}^2$ as a natural seed.

5.1 The equivalence

Definition 5.1. A pure n -dimensional complex K is *balanced* if there is a map $\mathfrak{c} : V \rightarrow \{0, \dots, n\}$, a *coloring*, that is injective on the vertices of every cell. Equivalently, the 1-skeleton is properly $(n+1)$ -colorable.

Proposition 5.2. *Let K be an n -dimensional conforming mesh. There is a choice of vertex order in every cell making K strongly compatible if and only if K is balanced. In that case the order of each cell is the order of the colors of its vertices, up to a global permutation of the color set.*

Proof. ($\Leftarrow$) Order each cell by increasing color. Then $\sigma[k]$ has color k . If $\partial_i \sigma = \partial_j \omega$, the shared facet carries all colors except i as seen from σ , and all except j as seen from ω . Hence $i = j$.

($\Rightarrow$) Let $\partial_i \sigma = \partial_j \omega$ as ordered simplices. Every vertex of the shared facet then occupies the same position in σ and in ω . Fix a vertex v . Since $\text{st}(v, K)$ is strongly connected, any two cells containing v are joined by a chain of facet-adjacent cells, all containing v . The shared facets along the chain contain v , so the position of v is constant along it. Set $\mathfrak{c}(v)$ equal to this common position. It is injective on each cell, because positions within a cell are distinct. $\square$

Remark 5.3 (Index-only matching is not enough). It is tempting to read strong compatibility as a condition on *labels*: the omitted vertex has the same local index $i = j$ on both sides of every shared facet. That is, a proper $(n+1)$ -edge-coloring of the dual graph in which every cell sees every color once. This is strictly weaker, and it does not suffice.

In dimension 3, take $\sigma = \langle a, b, c, d \rangle$ and $\omega = \langle d, x, c, a \rangle$, sharing the facet $\{a, c, d\}$ at index 1 on both sides. Both have split edge $\{a, d\}$. After one bisection with new vertex m , consider $c_0 \sigma = \langle a, b, c, m \rangle$ and $c_1 \omega = \langle x, c, a, m \rangle$. They share $\{a, c, m\}$ at indices 1 and 0 respectively, while type = 2. This violates (MaubachA): the split edge $\{a, c\}$ of $c_0 \sigma$ lies in $c_1 \omega$, whose split edge is $\{x, a\}$.

By König’s edge-coloring theorem, every closed pseudomanifold with bipartite dual graph admits index-only matchings. Such matchings are therefore far more common than balance. For instance, Kühnel’s $\mathbb{CP}_9^2$ admits one, although it is not balanced (Proposition 5.9).

5.2 Classical constructions as corollaries

Corollary 5.4 ([43, Thm. 9]). *The canonical (staircase) triangulation of a product of simplices $\Delta^{m_1} \times \dots \times \Delta^{m_k}$ is balanced, with $\mathfrak{c}(v_L) = |L|$. Hence it is strongly compatible. Kuhn’s triangulation of the cube is the case $m_1 = \dots = m_k = 1$.*

Corollary 5.5 ([43, Thm. 10]). *The barycentric subdivision $B(K)$ of any mesh is balanced, with the color of the barycenter of δ equal to $\dim \delta$. Hence $B(K)$ is strongly compatible. It can be produced by ordered stellar subdivisions $(\delta, v_\delta)_{\dim \delta}$, applied in order of decreasing dimension.*

Proposition 5.6. *Let K_1, K_2 be balanced, of dimensions m_1, m_2 , with colorings $\mathbf{c}_1, \mathbf{c}_2$. The staircase triangulation of $K_1 \times K_2$, built prism by prism from monotone lattice paths, is balanced with coloring $\mathbf{c}(a, b) = \mathbf{c}_1(a) + \mathbf{c}_2(b)$.*

Proof. A cell of the prism $\sigma_1 \times \sigma_2$ is a monotone path $L_0, \dots, L_{m_1+m_2}$ in $\{0, \dots, m_1\} \times \{0, \dots, m_2\}$ with $|L_l| = l$. Its l -th vertex is $(\sigma_1[L_l^1], \sigma_2[L_l^2])$, of color $L_l^1 + L_l^2 = l$. The coloring is defined on vertices, so it is consistent across prisms. $\square$

Example 5.7. The octahedron (the boundary of the cross-polytope) is balanced: color $\pm e_k$ by k . By Proposition 5.6, the staircase triangulation of $S^2 \times S^2$ as octahedron $\times$ octahedron is balanced. This is the seed used by the $S^2 \times S^2$ variant of our Riemann-surface code.

Stevenson’s pre-refinement [53, App. A], after Kossaczky [34], is essentially the barycentric subdivision without the edge-midpoint level. It produces $\frac{1}{2}(n+1)!$ cells per simplex instead of $(n+1)!$, and satisfies the weaker matching condition. The coloring initialization of [19] avoids pre-refinement altogether. For balanced meshes it coincides with Proposition 5.2; for others it uses $N > n$ colors.

5.3 Colorability criteria

Joswig [32] shows that a locally strongly connected complex is balanced iff its group of projectivities is trivial. For simply connected manifolds this reduces to Traxler’s criterion [54]: every $(n-2)$ -face has even degree. A cruder obstruction is often enough. A 2-neighborly complex with more than $n+1$ vertices is never balanced, because its graph is complete.

Lemma 5.8. *Let K be an orientable, strongly connected, balanced closed pseudomanifold. Then its dual graph is bipartite.*

Proof. Fix a coherent orientation. Order each cell by color and let $\varepsilon(\sigma) = \pm 1$ record whether this ordering is positively oriented. Adjacent cells σ, ω miss the same color k on their common facet F , and induce on F the orientations $(-1)^k \varepsilon(\sigma)[F]$ and $(-1)^k \varepsilon(\omega)[F]$, where $[F]$ is the color order of F . Coherence requires these to be opposite, so $\varepsilon(\sigma) = -\varepsilon(\omega)$. $\square$

5.4 Closed-manifold seeds: $\mathbb{CP}^2$

$\mathbb{CP}^2$ has two classical small triangulations. Kühnel’s $\mathbb{CP}_9^2$ [36] has 9 vertices and 36 cells and is vertex-minimal. Gaifullin’s [26] has 15 vertices and 108 cells, f-vector $(15, 90, 240, 270, 108)$, and automorphism group $S_4 \times S_3$ acting by Fubini–Study isometries in Gaifullin’s explicit realization.

Proposition 5.9. *Kühnel’s $\mathbb{CP}_9^2$ is not balanced.*

Proof. $\mathbb{CP}_9^2$ is 3-neighborly, so its graph is K_9 , which is not 5-colorable. $\square$

Theorem 5.10. *Gaifullin’s triangulation X of $\mathbb{CP}^2$ is balanced. Its five color classes are the natural blocks of its vertex set $(V_4 \setminus \{e\}) \sqcup (\{1, 2, 3, 4\} \times \{1, 2, 3\})$, namely $V_4 \setminus \{e\}$ and $\{a\} \times \{1, 2, 3\}$ for $a = 1, \dots, 4$. Consequently, X is a vertex-minimal balanced triangulation of $\mathbb{CP}^2$.*

Proof. Each block consists of three pairwise non-adjacent vertices. Every vertex has degree 12 in the 1-skeleton, which has 90 edges, and its 2 non-neighbors are exactly the other members of its block. So the complement of the 1-skeleton is the disjoint union of five triangles. Consequently every 4-simplex, having 5 pairwise adjacent vertices, meets each block exactly once. We verified this by direct computation on the facet list of [26], cross-checked against the simpcomp

library [24]. As an independent check, every one of the 240 triangles lies in 4 or 6 cells, as Traxler’s parity criterion requires.

For minimality, $\mathbb{CP}^2$ is orientable, so by Lemma 5.8 every balanced triangulation of $\mathbb{CP}^2$ admits a chess colouring of its 4-simplices. Gaifullin proves that X has the minimal number of vertices among triangulations of $\mathbb{CP}^2$ admitting a chess colouring [26]. $\square$

Remark 5.11. With the color order, all 270 interior facets of X satisfy $i = j$, and the 108 seed cells are geometrically uniform. The max/min Fubini–Study edge-length ratio is 1.4636 in every cell, as expected from the isometric symmetry. A seed obtained from Kühnel’s triangulation by barycentric subdivision (Corollary 5.5) is balanced as well. In Kühnel’s standard realization, however, several flag barycenters coincide with existing vertices, producing degenerate cells. Gaifullin’s triangulation avoids both problems without any subdivision.

6 Pointerless codes on balanced seeds

6.1 Why codes: the cost of explicit incidence

An explicit mesh data structure stores every simplex of every dimension with its boundary and coboundary references. Examples are the half-edge and its higher-dimensional analogues, incidence graphs, and the `nmt<d>` container of VGTL [42] used in our first implementation. The number of faces a cell carries grows quickly with d . In the Freudenthal–Kuhn triangulation of $\mathbb{R}^d$, which describes the local structure of any Maubach mesh, a k -face is determined up to translation by an ordered sequence of k disjoint nonempty subsets of $\{1, \dots, d\}$. There are therefore $k! S(d+1, k+1)$ of them per lattice point, where S denotes Stirling numbers of the second kind, and $k! S(d+1, k+1)/d!$ per d -cell. Storing only the boundary references ($k+1$ integers per k -face) already costs:

d	2	3	4	5	6	7	8
faces of dim. 1.. d per cell	2.5	4.2	6.2	9.0	13.0	18.8	27.1
boundary references per cell	6.0	12.3	22.5	39.0	65.7	108.3	175.8
LPT code bits (40 levels)	49	52	55	59	62	66	70

The explicit cost grows roughly by a factor 1.6 per dimension. Coboundary references, hierarchy links and per-simplex attributes come on top of it. The LPT code of a cell refined to L levels needs about $L + \log_2(2^d d!) + O(\log d)$ bits, regardless of how many faces the cell has.

A second cost is history. Maubach bisection replaces cells, and with them their faces. An append-only container (`nmt<4>` never frees replaced simplices, since other simplices may still refer to them) grows with the number of cells ever created, not with the current mesh.

We measured both effects on a pure-topology benchmark: uniform Maubach refinement of Gaifullin’s 108-cell seed of $\mathbb{CP}^2$ to depth D , with no coordinates and no application data. Both structures hold exactly the same $108 \cdot 2^D$ leaves.

D (leaves)	<code>nmt<4></code>		<code>glpt</code>	
	cells stored	bytes/leaf	table	bytes/leaf
8 (27 648)	55 188	222	0.5 MB	19.5
12 (442 368)	884 628	222	8.6 MB	19.5

The explicit structure stores twice as many cells as are current. It also stores 2.7 triangles and 4.0 tetrahedra per current cell, for 222 bytes of records per leaf. The pointerless table stores one 64-bit code per leaf, at load factor 0.41, for 19.5 bytes per leaf: an $11\times$ reduction before any application data is attached.

In the full application, per-simplex attributes such as coordinates and root caches multiply the explicit cost. The `nmt<4>`-based implementation peaked at 4–5 GB around refinement depth

18 on a machine with 7.2 GB of RAM, which made deeper runs impossible. The `glpt`-based one runs at depth 20 (Section 7).

6.2 LPT codes

Atalay and Mount [5] encode the Maubach hierarchy of a Kuhn-triangulated cube. Work in the cube $[-1, 1]^d$ with the reference root simplex

$$S_{\varnothing} = \{x \in \mathbb{R}^d : 1 \geq x_1 \geq x_2 \geq \cdots \geq x_d \geq -1\}.$$

Its facet H_0 lies on $x_1 = 1$, its facet H_j ($0 < j < d$) on $x_j = x_{j+1}$, and its facet H_d on $x_d = -1$. A descendant is encoded by its *LPT code* (ℓ, Π, Φ) :

- ℓ is its level modulo d ;
- $\Pi = (\pi_1, \dots, \pi_d)$ is a signed permutation;
- Φ is the *orthant list*. After every d levels, one bit is appended to each coordinate, recording in which half of the current cube the cell lies.

We read the orthant list as integers $o_1, \dots, o_d \in [0, 2^r)$, where r is the number of completed rounds of d levels and the oldest bit is the most significant.

Their Theorem 1 [5] gives the vertices explicitly. Let $T_j = 2^r - 1 - 2o_j$, so that $T/2^r$ is the center of the cell's cube. The k -th vertex is

$$x^{(k)} = \frac{T + y^{(k)}}{2^r}, \quad y_{|\pi_j|}^{(k)} = \text{sgn}(\pi_j) v_{kj}, \quad v_{kj} = \begin{cases} 1, & k \geq j, \\ 0, & k < \min(j, \ell), \\ -1, & \ell \leq k < j, \end{cases} \quad (2)$$

for $k = 0, \dots, d$ and $j = 1, \dots, d$. The vectors $y^{(k)} \in \{-1, 0, 1\}^d$ depend only on (ℓ, Π, k) . Only the center T depends on the orthant list. Their Theorem 2 [5] computes the code of a facet neighbor by table lookup.

Children, parents and neighbors are thus pure integer operations on a machine word. Our implementation descends from the one written for the Adaptive Voids algorithm [23]. There, LPT codes over the six Kuhn tetrahedra of a cube drive an adaptive tetrahedral subdivision of a solid, whose cells are then shrunk into printable voids. A mesh is just the set of codes of its current leaves. Atalay and Mount's Theorem 2 describes neighbors in the *infinite* Kuhn arrangement. For the whole cube, which is the union of the $d!2^d$ images of $S_{\varnothing}$, a neighbor outside one root simplex is always another root simplex of the same cube, reached by composing signed permutations.

6.3 Balanced seeds

Let K_0 be a balanced d -dimensional seed with N_0 cells, each ordered by color. Identify every seed cell $\tau = \langle \tau[0], \dots, \tau[d] \rangle$ with $S_{\varnothing}$ through the barycentric map sending the k -th vertex of $S_{\varnothing}$ to $\tau[k]$. A `glpt` [41] cell is a pair (s, κ) : a seed index s and an LPT code κ of the reference hierarchy. By (2), the m -th vertex of (s, κ) is the point with barycentric coordinates $w(\kappa, m) \in \Delta^d$ with respect to τ_s . These coordinates are dyadic rationals determined by κ alone. Two questions remain: when does a facet leave its seed cell, and where does it land?

Proposition 6.1 (seed-boundary test). *Facet ∂_i of the cell with code $\kappa = (\ell, \Pi, \Phi)$ lies on the boundary facet H_j of its seed cell if and only if both of the following hold:*

- (i) *the linear part of the equation of H_j is constant on $\{y^{(k)} : k \neq i\}$, i.e. $y_1^{(k)} \equiv 1$ for $j = 0$, $y_{j+1}^{(k)} - y_j^{(k)} \equiv 2\delta$ with $\delta \in \{-1, 0, 1\}$ for $0 < j < d$, and $y_d^{(k)} \equiv -1$ for $j = d$;*

- (ii) the orthant coordinates satisfy $o_1 = 0$ for $j = 0$, $o_{j+1} - o_j = \delta$ for $0 < j < d$, and $o_d = 2^r - 1$ for $j = d$.

At most one j satisfies (i). Condition (i) depends only on (ℓ, Π, i) .

Proof. Substitute (2) into the equation of H_j . For $0 < j < d$ the equation reads $T_j + y_j^{(k)} = T_{j+1} + y_{j+1}^{(k)}$ for all $k \neq i$. Since T does not depend on k , this forces $y_{j+1}^{(k)} - y_j^{(k)}$ to be constant in k , equal to $T_j - T_{j+1} = 2(o_{j+1} - o_j)$. For $j = 0$, the equation $T_1 + y_1^{(k)} = 2^r$ gives $2o_1 = y_1^{(k)} - 1$. Since $o_1 \geq 0$ and $y_1^{(k)} \leq 1$, this means $o_1 = 0$ and $y_1^{(k)} = 1$. The case $j = d$ is symmetric. For uniqueness, note that the d vertices of a facet span a hyperplane, which has a unique normal direction. The normals of the $d + 1$ facets of a simplex are pairwise non-parallel. $\square$

Condition (i) is precomputed once, in a table indexed by (ℓ, Π, i) that stores j and δ , or “none”. For $d = 4$ the table has $4 \cdot 384 \cdot 5 = 7680$ one-byte entries. The test is then a single lookup followed by a single integer comparison. There is no arithmetic on coordinates, no loop over refinement rounds and no floating point.

Proposition 6.2 (seed crossing). *Let K_0 be balanced and ordered by color. Suppose facet ∂_i of $c = (s, \kappa)$ lies on the boundary facet $\partial_j \tau_s$. Let s' be the seed cell sharing that facet. Then the neighbor of c across $\partial_i c$ is $c' = (s', \kappa)$, the same code in the other seed cell, and $\partial_i c' = \partial_i c$ as ordered simplices.*

Proof. By Proposition 5.2, $\partial_j \tau_s = \partial_j \tau_{s'}$ as ordered simplices, so $\tau_s[m] = \tau_{s'}[m]$ for every $m \neq j$. For $m \neq i$, the m -th vertex of c lies on $\partial_j \tau_s$, so $w_j(\kappa, m) = 0$. It is therefore $\sum_{p \neq j} w_p(\kappa, m) \tau_s[p] = \sum_{p \neq j} w_p(\kappa, m) \tau_{s'}[p]$, which is the m -th vertex of c' . Hence c' contains $\partial_i c$ in the same positions. It has the same level as c and lies in the other seed cell. Uniform refinements of a strongly compatible mesh are conforming (a consequence of Theorem 4.5; also [54, 53]), so exactly two cells of that level contain $\partial_i c$. Therefore c' is the neighbor. $\square$

Proposition 6.2 is where balance, rather than a mere compatible ordering, pays off. Without it, the two identifications would disagree on the shared facet by a permutation, and the code would have to be transformed.

6.4 Neighbor finding

The resulting neighbor procedure is entirely combinatorial. To find the neighbor of $c = (s, \kappa)$ across $\partial_i c$:

1. *Sibling facet.* If $\partial_i c$ is the facet created by the last bisection ($i = d$ for a 0-child, $i = \text{type}$ for a 1-child), return the other child of the parent (Lemma 4.3).
2. *Boundary test.* Otherwise, look up (j, δ) for (ℓ, Π, i) and check the orthant condition of Proposition 6.1.
3. *Interior facet.* If the test fails, return (s, κ') , where κ' is given by Atalay and Mount’s Theorem 2.
4. *Seed crossing.* If it succeeds, return $(\text{adj}[s][j], \kappa)$ (Proposition 6.2). The seed adjacency table has $N_0 \times (d+1)$ entries; a root cell uses it directly with $j = i$.

Figure 1 illustrates rules 2 and 4 in dimension 3. Rule 2 is indispensable. Atalay and Mount’s Theorem 2 describes neighbors in the infinite Kuhn arrangement, so across a seed-boundary facet it returns a cell of the reflected reference simplex, i.e. a cell with negative barycentric weights. Such crossings are frequent: about 24% of the non-sibling facets sampled at realistic depths lie on the seed boundary.

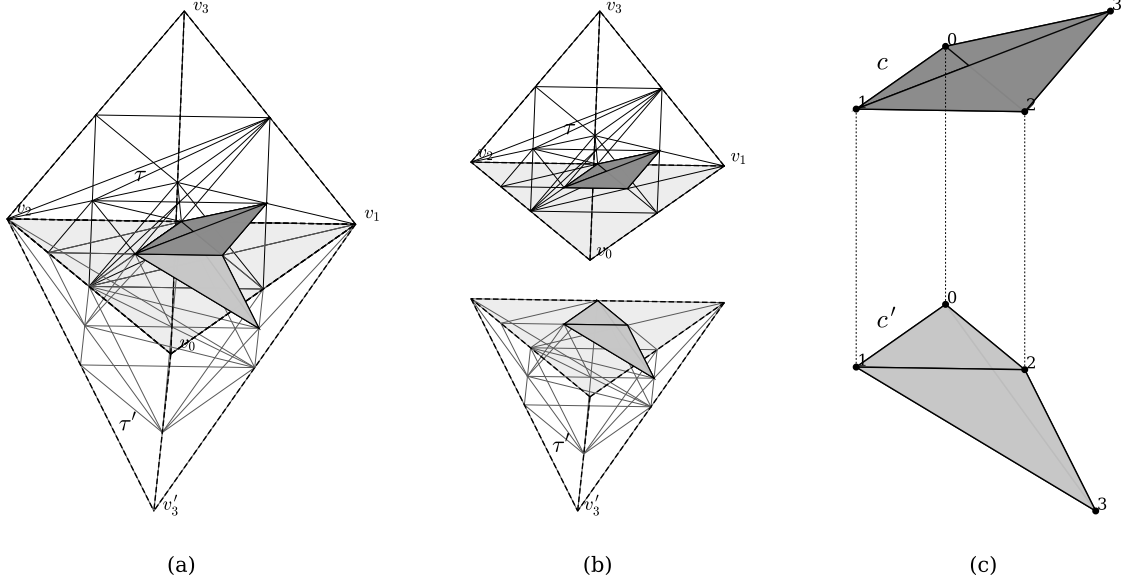


Figure 1: Propositions 6.1 and 6.2 in dimension 3. (a) Two colour-ordered seed tetrahedra τ and τ' , glued along their common facet $\langle v_0, v_1, v_2 \rangle$ (light grey), after adaptive Maubach bisection towards a point just above that facet. The conforming closure propagates across the seed boundary. Dashed lines are the seed edges. (b) The same mesh, exploded along the normal of the common facet: both seed cells induce the same subdivision on it. (c) The highlighted leaf $c \subset \tau$, whose facet $\partial_3 c$ lies on the seed boundary, and its neighbor $c' \subset \tau'$. Both have the same bisection code (path 010010 from their roots), and the shared facet occupies local positions 0, 1, 2 in both. The figure is generated by a script that asserts both facts.

Verification. Every returned neighbor was checked *exactly*. Both cells were written as integer barycentric weights over the 15 global vertices of Gaifullin’s triangulation, with common denominator 2^{11} . We then verified that the neighbor is a different cell of the same level containing the queried facet. The check covered 10^6 random cells over the whole representable depth range, 0 to 43, and 5 million neighbor queries:

- sibling: 977 070 queries;
- interior: 3 317 979 queries;
- seed crossing: 704 951 queries.

No query failed. Away from sibling facets, the facet always sits at the *same* local index in both cells, as strong compatibility predicts. Orientation coherence was checked on 750 000 sampled (cell, facet) pairs, with no violation.

Comparison with the first implementation. Our first version used a different boundary test. It computed the full matrix of exact barycentric weights of the facet’s vertices, which costs $O(d^2 + rd)$ integer operations. It also resolved crossings geometrically: a point slightly across the facet was located by descending the neighboring seed’s hierarchy in floating point. On 7.8 million sampled facets, the table-based test of Proposition 6.1 agrees with the weight-matrix test in every case, and the same-code rule agrees with geometric location on all 1.4 million crossings. Both are also faster: 4.3 ns against 79 ns per boundary test, and 5 ns against 211 ns per crossing. The earlier floating-point path was the only source of discrepancies in the first implementation, at the maximum representable depth; it is now gone. The application of Section 7 produces byte-identical output with either implementation. Its running time is dominated by root finding on 2-faces, so the end-to-end speedup is small, about 2%.

6.5 The leaf set and graded conforming refinement

The mesh is the set of codes of the current leaves. They are stored in an open-addressing hash table with one 64-bit slot per leaf, linear probing and backward-shift deletion; the last avoids tombstones under the constant erase-two-insert churn of bisection. For Gaifullin’s seed ($d = 4$, $N_0 = 108$), a code fits in one word:

field	seed	ℓ	Π	r	$o_1, \dots, o_4$	total
bits	7	2	9	4	4×10	62

This allows $r \leq 10$ rounds, i.e. depth 43.

Two operations sit on top of this table.

- `compat_bisect( $c$ )` is Algorithm 1 expressed on codes. Before bisecting c , it recursively bisects coarser neighbors, so that the mesh stays graded: no facet separates cells more than one level apart.
- `neighbor_leaf( $c, i$ )` walks up the ancestor chain of the combinatorial neighbor until it finds a current leaf. If the neighbor has been refined further than c , it reports this instead.

An exhaustive graded-invariant test checked about 40 000 cascading bisections producing roughly 4 million leaves, 11.6 million facet checks in total, with no violation.

Vertex identities and coordinates are not part of the code. When a cell is created, its new vertex receives a global integer identifier, keyed by the identifiers of the endpoints of the split edge in a separate hash table. Coordinates are attached to vertex identifiers only. Vertices are far fewer than cells: about $1/d!$ per cell asymptotically.

Limitations. `glpt` does not yet enumerate the several finer leaves that may border a coarser facet. Such queries are reported, never guessed. For seeds that are compatible only in the weak sense of (MaubachA), with the $\{0, \text{type}\}$ exception, the two identifications disagree on shared facets, and Proposition 6.2 would need a permutation correction. For seeds initialized as in [19] with $N > d$ colors, a cell is a face of a virtual N -simplex. Its code would live in the N -dimensional reference hierarchy, but [19, Rem. 12] shows that refinements of the virtual extension do not restrict to refinements of the mesh. We leave both cases open.

7 Application: Riemann surfaces in $\mathbb{CP}^2$

7.1 Setting

Let $F(X, Y, Z)$ be a homogeneous polynomial of degree n defining a smooth curve $C = \{F = 0\} \subset \mathbb{CP}^2$. C is a compact Riemann surface of genus $(n-1)(n-2)/2$, a real surface of codimension 2 in the real 4-manifold $\mathbb{CP}^2$. We triangulate C in three steps:

1. mesh $\mathbb{CP}^2$ adaptively, starting from Gaifullin’s triangulation realized with its explicit coordinates [26];
2. refine by pointerless Maubach bisection (Section 6), placing each new vertex at the *Fubini–Study geodesic midpoint* of the split edge;
3. extract C cell by cell.

The resulting cells are not affine simplices in any chart. Their vertices are points of $\mathbb{CP}^2$ and their 2-faces are handled in the best-conditioned affine chart of each face. Midpoints are computed after aligning the phases of the two homogeneous representatives. Because the geodesic midpoint

Algorithm 2 Continuation triangulation of $C = \{F = 0\}$

Require: balanced seed T (as a `glpt` leaf set); curve F ; target depth D

```
1:  $s \leftarrow$  first seed cell whose extraction finds a crossing
2:  $visited \leftarrow \emptyset$ ;  $frontier \leftarrow \{s\}$ 
3: while  $frontier \neq \emptyset$  do
4:   pop  $c$ ; skip if  $c \notin T$  or  $c \in visited$ 
5:    $visited \leftarrow visited \cup \{c\}$ ;  $r \leftarrow \text{extract}(c)$ 
6:   skip if  $r$  has no crossing
7:   if  $\text{level}(c) < D$  then
8:      $T.\text{compat\_bisect}(c)$ ; add the newly created leaves to  $frontier$ 
9:   else
10:    record  $r$ 
11:    for  $i = 0, \dots, 4$  do
12:      if  $T.\text{neighbor\_leaf}(c, i)$  finds  $n \notin visited$  then add  $n$  to  $frontier$ 
13:    end for
14:  end if
15: end while
```

does not commute with linear combinations, vertex coordinates are threaded along the bisection path from the root, cached by global vertex identifiers.

To avoid resonance between the curve and the seed's coordinate axes, the seed may be moved by a random unitary change of basis of $\mathbb{C}^3$.

7.2 Per-cell extraction

For a leaf c , each of its 10 triangular 2-faces is intersected with C by Newton's method in barycentric coordinates, from several seeds. Solutions are cached by the sorted triple of global vertex identifiers, so that neighboring cells see identical crossing points. Generically C meets a 2-face transversally in isolated points. The crossing points of the 5 tetrahedral facets of c are paired into arcs (by minimal total geodesic length when a facet has four crossings). The resulting graph is decomposed into closed polygons approximating $C \cap c$. A cell is *bad* when this decomposition fails: a node of degree $\neq 2$, or three crossings on one facet.

An optional mode encloses F on a 2-face using its Bernstein–Bézier coefficients on the triangle, in the spirit of [50]. It is used to certify empty faces and to recover roots missed by Newton.

7.3 Continuation

Algorithm 2 follows the curve from a single cell it is known to meet. Only the combinatorial operations of Section 6 are used to move through the mesh.

If C crosses a facet shared by cells A and B , it also meets B , so propagation through crossing facets reaches every cell met by the connected curve. Reducible curves need one seed per component.

Two refinement mechanisms act beyond the target depth.

- *Proactive: tangency threshold.* At each crossing, let w_k be the complex directional derivatives of F along the two edge vectors e_1, e_2 of the 2-face, and (F_a, F_b) its chart gradient. By the Cauchy–Riemann equations, the quantity

$$\text{score} = \frac{|\text{Im}(\overline{w_1}w_2)|}{|e_1||e_2|(|F_a|^2 + |F_b|^2)} \in [0, 1]$$

vanishes exactly when the face is tangent to C at the crossing. Cells whose minimal score falls below a threshold are bisected further, within an extra depth budget.

- *Reactive: repair rounds.* Bad cells are re-bisected and re-extracted.

7.4 Results

We tested six smooth curves: a conic, the Fermat cubic, a generic Hesse-pencil cubic, a smooth quartic, a parabola and an elliptic curve. Continuation recovers exactly the same leaf set and the same bad cells as a global scan of all 108 seed hierarchies with the same test, at depths 8, 12 and 14. Compared with the earlier refinement criterion based on gradient dispersion, which refines all of $\mathbb{CP}^2$, continuation yields meshes 3 to 6 times smaller at depth 14. Its bad-cell rate is never worse.

For the elliptic curve $X^2Z - Y^3 + YZ^2 = 0$, with a random unitary rotation of the seed:

depth	mode	leaves	good cells	bad cells	bad rate
14	baseline	327 370	27 397	887	3.14%
14	repair rounds (2)	327 370	30 605	730	2.33%
14	tangency threshold 0.05	612 978	42 278	904	2.09%
14	both	612 978	45 709	729	1.57%
20	baseline	4 620 102	222 981	1 348	0.601%
20	repair rounds (2)	4 678 426	226 677	678	0.298%
20	tangency threshold 0.05	6 788 844	335 517	1 260	0.374%
20	both	6 842 524	339 147	610	0.180%

At depth 20 (baseline), we evaluated $|F|$ at the 164 390 distinct extracted vertices, with unit-norm homogeneous representatives. The median is 1.0×10^{-16} , the 99th percentile 4.3×10^{-13} and the maximum 9.8×10^{-13} .

Instrumenting the extraction at depth 14 (136 786 touched cells over all six curves, 8.0% bad) attributes the failures to near-tangency:

- 82% of the bad cells fail on node degree, 18% on facets with three crossings;
- cells containing a facet touched at a single point are 7 times more likely to be bad;
- the tangency score separates good cells (median 0.21) from bad ones (median 0.12), with a monotone gradient in bad rate across score buckets.

Bisection with Fubini–Study midpoints splits cell volumes unevenly, but this imbalance does *not* correlate with bad cells.

Figure 2 shows two of the resulting surfaces.

Topological check. Although the output is not certified, its topology can be checked a posteriori, directly on the extracted complex in $\mathbb{CP}^2$ and independently of any projection. Output vertices are identified combinatorially: a crossing is keyed by the mesh vertex, edge or 2-face it lies on. The extracted polygons therefore form a single 2-complex. Each bad cell leaves a hole. A hole is a loop of edges that belong to a single polygon. A vertex is *pinched* if its link has more than one component; we split every pinched vertex into one copy per link component. In all our runs, every pinched vertex lies on hole boundaries: its link consists of exactly two paths. About two thirds are points where the outline of a single hole touches itself; the rest are points where two holes meet. We never observed two sheets touching in the interior. If the remaining holes are discs, the resulting surface with c components and b boundary loops satisfies

$$\chi = \sum_{i=1}^c (2 - 2g_i) - b, \quad \text{hence} \quad g = \sum_i g_i = \frac{2c - b - \chi}{2}.$$

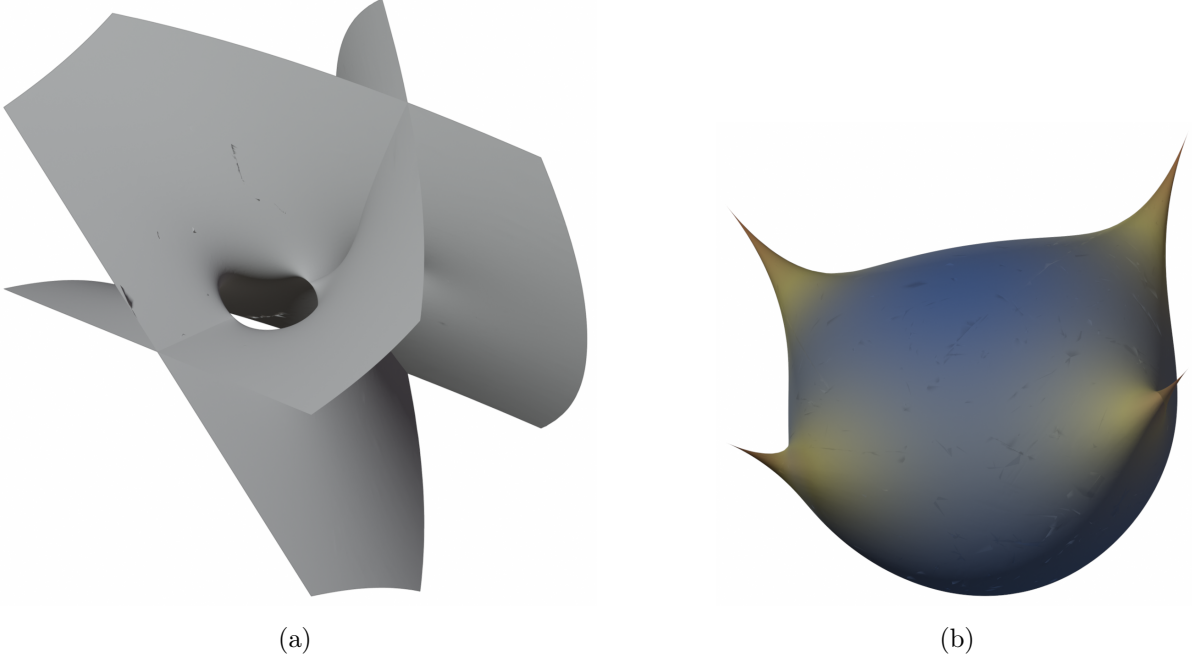


Figure 2: Riemann surfaces triangulated intrinsically in $\mathbb{CP}^2$ by Algorithm 2. (a) The elliptic curve $X^2Z - Y^3 + YZ^2 = 0$, i.e. $w^2 = z^3 - z$ in the chart $Z = 1$ with $w = X$ and $z = Y$, shown as the classical graph $(\operatorname{Re} z, \operatorname{Im} z, \operatorname{Re} w)$, clipped to $|\operatorname{Re} z|, |\operatorname{Im} z| \leq 2$ and $|\operatorname{Re} w| \leq 3$. The two sheets meet over the branch points $z = 0, \pm 1$. Depth 18 with tangency threshold 0.05 and two repair rounds: 3.03M leaves, 177 650 good and 701 bad cells (0.39%). (b) The Fermat quartic $X^4 + Y^4 + Z^4 = 0$ (genus 3) under Dutter's chart-free projection $\tilde{\alpha} : \mathbb{CP}^2 \rightarrow \mathbb{R}^3$ [22]. It is coloured by distance to the centre, from blue (near) to red (far), and drawn semi-transparent. Depth 17, same settings: 2.40M leaves, 155 436 good and 5 388 bad cells (3.35%). Both surfaces were closed by the topological post-processing described in Section 7: cones over the holes left by bad cells, with apices on the curve. The closed complexes were verified to be closed surfaces of the correct genus. The faint marks visible on both surfaces are discontinuities of the interpolated normals, not holes. They occur where the small closing cones meet the surrounding mesh (most of the marks in (b)), and at a few thin or slightly folded polygons produced by the extraction itself (the darker streaks in (a)). Rendering scripts are provided with the code [42].

The estimate recovers the genus $(n-1)(n-2)/2$ exactly in every case we tried, even at a bad-cell rate close to 6%:

curve	depth	bad cells	pinched	c	b	χ	g est.	g
conic $XY - Z^2$	14	367 (1.2%)	3	2	41	-37	0	0
elliptic $X^2Z - Y^3 + YZ^2$	14	729 (1.6%)	8	4	84	-78	1	1
elliptic, Fig. 2(a)	18	701 (0.39%)	3	2	104	-102	1	1
Fermat cubic	14	2 438 (5.9%)	35	8	253	-239	1	1
Fermat quartic	17	5 388 (3.4%)	36	11	656	-640	3	3

All runs use a random unitary rotation of the seed, tangency threshold 0.05 and two repair rounds. That the check succeeds shows that the failures are local: each hole is a disc, and no hole cuts through a handle of the surface.

Topological closing. The same analysis yields a simple post-processing step that turns the extracted complex into a closed surface of the correct topology:

1. split the pinched vertices as above;

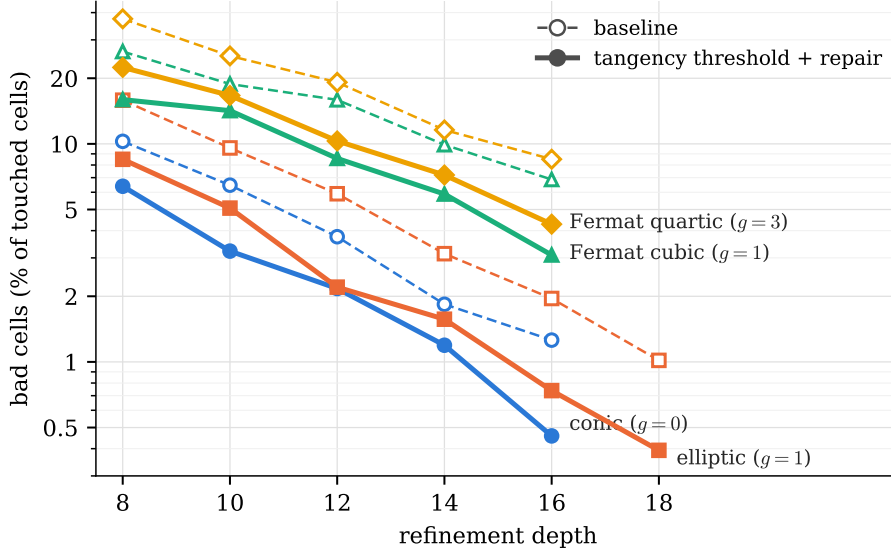


Figure 3: Percentage of touched cells whose extraction fails, against the refinement depth, for four smooth curves (seed rotated by the same random unitary change of basis). Dashed lines with open markers: continuation alone. Solid lines: with tangency threshold 0.05 and two repair rounds. The elliptic curve is also shown at depth 18, the mesh of Figure 2(a). Data and scripts are provided with the code [42].

2. keep the main connected component. The splitting cuts off a few tiny islands, from 1 to 18 faces in our runs, which lay inside a hole whose outline touched itself;
3. cap every boundary loop by a cone from a *new* apex vertex. A fan from an existing vertex of the loop could duplicate an edge that already exists in the mesh. The apex is placed at the Fubini–Study barycentre of the loop and moved onto the curve by Newton’s method in the best affine chart (minimum-norm steps for one complex equation in $\mathbb{C}^2$);
4. verify the result: every edge lies in exactly two faces, every vertex link is a single cycle, the complex is connected, and $\chi = 2 - 2g$ with $g = (n - 1)(n - 2)/2$.

The last step makes the procedure self-certifying with respect to topology: a hole that was not a disc would change χ and be detected. On all the runs of the previous table, the closed surface passes every check: the conic gives $\chi = 2$, the cubics $\chi = 0$, and the quartic $\chi = -4$. In every case, Newton’s method moved every cone apex onto the curve. Being topological, the procedure acts on the complex in $\mathbb{CP}^2$ and is independent of the projection later used for display. The surfaces of Figure 2 are closed this way.

Bad-cell rate versus depth. Figure 3 shows how the bad-cell rate evolves with the refinement depth for four curves of degree 2 to 4. The rate decreases steadily, roughly halving every two levels. The proactive tangency threshold combined with repair rounds lowers it by a factor of 1.5 to 3 at every depth. Curves of higher degree need a few more levels to reach the same rate.

7.5 Discussion

Four features distinguish the construction from chart-based approaches.

- *Intrinsic.* $\mathbb{CP}^2$ is compact and the seed is symmetric under a group of isometries, so there are no charts, branch cuts or special points at infinity. Local affine charts appear only inside the numerical solve on a single 2-face.

- *Codimension two.* The extraction runs directly in the real 4-manifold, as in [58], but on an adaptive mesh.
- *Combinatorial navigation.* The neighbor structure of the pointerless mesh is all the continuation needs. Memory is proportional to the number of leaves, which allowed depth 20 with close to 7 million leaves on a workstation with 7.2 GB of RAM.
- *Not certified.* Unlike [49, 12], the output is not guaranteed to be isotopic to C . The residual bad cells, concentrated at near-tangencies, are the price. Their effect is purely local, however: after topological closing, the result is a closed surface whose genus is verified to be correct.

A certified variant would combine the Bernstein enclosures with a transversality test on each 2-face. We leave this for future work.

8 Conclusion

Stellar subdivision schemes reduce the correctness of Maubach bisection to three local lemmas about ordered stellar operations. On conforming meshes, the strong form of the invariant is precisely balance. This single notion explains:

- the classical constructions (products of simplices and barycentric subdivisions);
- the coloring-based initializations of the finite-element literature;
- the portability of pointerless LPT codes to arbitrary seeds.

Balanced triangulations of closed manifolds become natural seeds. For $\mathbb{CP}^2$, Gaifullin’s triangulation is a vertex-minimal balanced one. On top of it, pointerless Maubach bisection and combinatorial continuation triangulate Riemann surfaces intrinsically in $\mathbb{CP}^2$.

Several questions remain open:

- a precise dictionary between (MaubachA) and Stevenson’s matching condition;
- pointerless codes for weakly compatible seeds, and for the $N > n$ initializations of [19];
- a systematic search for small balanced triangulations of other closed manifolds, for example with the tools of [57];
- Gaifullin’s 33-vertex subdivision, on which the moment map is simplicial, as an alternative seed;
- certification of the extraction.

Note on AI assistance

The implementation described here was developed with Claude (Anthropic) acting as a coding assistant, under the author’s direction, review and testing. The software comprises the `glpt` pointerless mesh, the continuation algorithm and the diagnostic measurements. This draft was likewise prepared with Claude’s assistance.

Code availability. The ordered-mesh data structures and the stellar subdivision algorithms, including the Mitchell and Maubach schemes and the Riemann-surface programs of Section 7, are part of VGTL [42]. The LPT codes and their `glpt` extension to balanced seeds, with the test suites used in Section 6, are in the `lpt` repository [41].

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